MARITIME COMMAND

QUALIFICATION STANDARD AND PLAN

MARINE ENGINEERING TECHNICIAN (MOSID 00123)

HALIFAX CLASS DIESEL ENGINE MAINTAINER

Qualification Code: 108759

RECORD OF AMENDMENTS

The "Authority" reference indicates approval of the corresponding chapters of this QSP document. Authority for Chapters 1 and 2 of this QSP resides with CMS/DMTE. The Curriculum Control Authority for the training is the Canadian Forces Naval Engineering School Halifax (CFNESH). The latest approval date entries from CMS and the Training Establishment (CCA) should appear in the bottom right corner of every page.

REVISION	APPROVAL DATE	AUTHORITY

Chapters 1,2:
Chapters 3,4:

Chapters 1,2:
Chapters 3,4:

TABLE OF CONTENTS

Record of Amendments
Table of Contents
Foreword
CHAPTER 1: TRAINING MANAGEMENT DETAILS
- AIM
- COURSE LENGTH CALCULATION
- RESOURCES
- COURSE CAPACITY
- COURSE PREREQUISITES
- RELATED DOCUMENTS
- EVALUATION
- VALIDATION
- QUALIFICATION
- PRIOR LEARNING ASSESSMENT
- SPECIAL INSTRUCTIONS

CHAPTER 2: PERFORMANCE OBJECTIVES
- PO 401
- Maintain HALIFAX Class Diesel Engines
- Annex A – Main References
- Annex B – Task List

CHAPTER 3: ASSESSMENT OF TRAINEE
- GENERAL
- PERFORMANCE CHECKS
- PROGRESS MONITORING
- UNSATISFACTORY TRAINING PROGRESS
- COURSE REPORTS
- TRAINING REVIEW BOARDS

CHAPTER 4: LESSON SPECIFICATIONS
- EO 401.1
- Identify MWM Diesel Engine Components
- EO 401.2
- Identify MWM Diesel Engine Sub-Systems
- EO 401.3
- Perform MWM Diesel Engine Maintenance
- EO 401.4
- Identify Pielstick Diesel Engine Components

EO 401.5
Identify Pielstick Diesel Engine Sub-Systems
EO 401.6
Perform Pielstick Diesel Engine Maintenance
EO 401.7
Troubleshoot Diesel Engines
ANNEX A – Main References
ANNEX B – Suggested Sequencing and Timings

FOREWORD

1. This training document is developed under the authority of the Chief Maritime Staff in accordance with the following references:

a. A-P9-050 Series, Canadian Forces Manual of Individual Training and Education; and

b. A-PD-055-003/PQ-001, OSS AIBX - Canadian Patrol Frigate (CPF) Diesel Engine Maintenance (last modified on 5 May 2003).

2. This publication is effective on receipt. Performance Objectives (POs) are listed in the Table of Contents and are explained fully in Chapter 2. The Lesson Specifications (LSs) / Enabling Objectives (EOs), and Performance Check (PC) which specify the standards that the trainees must achieve to become qualified are listed in Chapter 4.

3. Suggestions for amendments may be initiated by anyone. Amendments must be forwarded through the normal chain of command. Specifically, training establishments are to summarize all requested amendments in writing to the applicable approving authority (ies). Final approvals for amendments are as follows:

a. Chapters 1 and 2 – Chief Maritime Staff, Director Maritime Training and Education (DMTE). Suggested amendments must first have the approval of the Curriculum Control Authority Training Establishment; and

b. Chapters 3 and 4 – Commandant of the Canadian Forces Naval Engineering School Halifax (CFNESH). Training Establishments that conduct training and are not the CCA must forward suggested amendments to CFNESH for approval.

4.

CCA	Phase	Qualification
CFNESH	HALIFAX Class Diesel Engine Maintainer	108759

CHAPTER 1: TRAINING MANAGEMENT DETAILS

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY CHIEF MARITIME STAFF/ DMTE

AIM

1. The aim of this course is to prepare Marine Engineering Technicians and Marine Engineering Artificers to perform maintenance and fault diagnosis on the diesel engines fitted in HALIFAX Class ships (Canadian Patrol Frigates).

COURSE LENGTH CALCULATION

2. Course length calculation (IAW MARCORD 9-30):

Actual Training Days (7 x 50 minute periods per day)	13
TAT	N/A
Travel Time	2
Course Duration (Total)	15

RESOURCES

3. The following resources are required:

Dedicated Staff Required

TITLE	RANK	QUALIFICATION	QTY
Senior Diesel Instructor	PO1	AIBX, Pielstick and MWM OST	1
Diesel Instructor	PO2	AIBX, Pielstick and MWM OST	1

Critical Training Resources

	No of Days
Fully functional MWM 602 and static MWM 602 diesel engine	All
Static Pielstick diesel engine	All
Fully functional MWM 226 and static MWM 226 diesel engine	All
Maintenance tools	All
Large test cell with test equipment	All

4. Other requirements:

a. FMF governor shop for 1/2 day; and

b. fully equipped classroom for 8 trainees.

COURSE CAPACITY

5. Planning for the conduct of training is based on the following guidelines:

a. maximum - 8;
b. optimum - 6; and
c. minimum – 4. COURSE

PREREQUISITES

6. Candidates must be minimum:

a. Legacy MAR ENG or HULL TECH QL5 qualified with the rank of LS or above;

b. MAR TECH USQ-LS Mechanical Specialist with the rank of LS or above;

c. posted to, or likely to be posted to:

(1) a HALIFAX Class ship (CPF);

(2) the diesel section of a Training Establishment; or

(3) Fleet Maintenance Facility (FMF) CAPE SCOTT or CAPE BRETON.

RELATED DOCUMENTS

7. A-PD-055-002/PP-002, Canadian Forces Manual of Military Occupational Structure, Volume 2, Part 2 - Occupational Specification for the Marine Engineering Occupation, MOCs 313 and 314, dated 30 January 1998.

EVALUATION

8. This is a pass/fail course. Evaluation of trainees will be conducted in accordance with Chapters 3 and 4 of this document.

VALIDATION

9. The training will be validated every five years in accordance with Naval Training Instructions, based on the task list in Chapter 2, Annex B.

QUALIFICATION

10. Successful completion of this course constitutes the only means of attaining the HALIFAX Class (CPF) Diesel Engine Maintainer qualification (108759).

PRIOR LEARNING ASSESSMENT

11. N/A.

SPECIAL INSTRUCTIONS

12. N/A.

CHAPTER 2: PERFORMANCE OBJECTIVES

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY CHIEF MARITIME STAFF/DMTE

PO 401

1. Performance. Maintain HALIFAX Class Diesel Engines.

2. Conditions.

   a. Given:

      (1) operational MWM and Pielstick diesel engines;

      (2) tools and test equipment;

      (3) replacement parts;

      (4) personal protective equipment (PPE); and

(5) References.

b. Denied: Assistance.

c. Environmental: At sea or alongside in all weather conditions.

3. Standard. Using the appropriate tools and test equipment, and employing all safety procedures and regulations, correctly maintain HALIFAX Class diesel engines, including:

a. the MWM diesel engine; and

b. the Pielstick diesel engine.

4. References. A1, A2, A3, A4, B1, B2, C1.

5. Specification Task Numbers

1. Inspect CPF diesel engines.

2. Maintain CPF diesel engines.

6. Supporting Skills.

1. Assembling/disassembling diesel engine components.

2. Evaluating components.

3. Identifying components.

4. Rectifying faults.

5. Troubleshooting.

6. Using specialized tools.

7. Supporting Knowledge

1. CFTOs and technical publications.

2. Diesel engine sub-systems.

3. Diesel engine theory and operating parameters.

4. Maintenance procedures.

5. Tolerance and wear limits.

6. Use of special tools and procedures.

8. Training Limitations. Instructors must be qualified through out-of-service training. Since this instruction is held in France and Germany, it is difficult to ensure qualified instructors are present at all times in the Training Establishment.

ANNEX A – MAIN REFERENCES

REFERENCE	CONTROL NUMBER	TITLE
A1	C-26-411-000/MS-001	MWM Technical Manual
A2	C-26-411-000/MY-001	MWM Parts Manual
A3	C-24-548-000/MS-001	Pielstick Operation and Maintenance Manual

A4	C-24-548-000/MS-002	Pielstick Parts and Tooling Manual
A5	C-27-499-000/MS-001	MWM 226 Technical Manual
B1	Publication 6.226.0.006.7802	MWM 226 Repair Manual
B2	Publication 0297 5201 TD 226	Deutz Service and Operation Manual
C1	ISBN 0-87006-424-X	Bill Toboldt. Diesel Fundamentals, Service and Repair, 1983.

2-A-1 **ANNEX B – TASK LIST**

This Task List is based on the A-PD-055-003/PQ-001, OSS AIBX dated 25 June 1997 (and modified on 5 May 2003) and is to be used for validation purposes.

ITEM	TASK/TASK ELEMENT	TASK SERIAL	DISPOSITION TRG/NO TRG	PO NO
1	Inspect CPF diesel engines	1	Train	401
2	Maintain CPF diesel engines	2	Train	401

2-B-1

CHAPTER 3: ASSESSMENT OF TRAINEE

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY THE COMMANDANT OF THE CANADIAN FORCES NAVAL ENGINEERING SCHOOL HALIFAX (CFNESH)

GENERAL

1. Candidates must successfully complete all the requirements detailed in Chapter 4 of this QSP and achieve an overall pass mark of 60% for the course in order to achieve the qualification (108759).

2. This is a pass/fail course and the assessment of candidates will be based on the successful completion of the Performance Objectives noted in Chapter 2 of this document. Grades and ranking will not be assigned.

PERFORMANCE CHECKS

3. The pass mark for each examination is 60%. For practical PCs, checklists will be used and students will be expected to achieve all the critical items in order to be successful.

4. A candidate who fails a PC will be permitted one supplementary attempt and the instructor shall ensure that sufficient time is scheduled accordingly. A pass on the supplementary attempt will reflect a pass mark of 60%, irrespective of a higher mark achieved.

PROGRESS MONITORING

5. Progress monitoring will provide candidates with an assessment of their progress throughout the course and assistance shall be provided to any candidate who experiences difficulty. A candidate who experiences difficulty must be given every chance to improve; however, he/she must be informed of the consequences of failure or marginal assessment and the disposition options.

UNSATISFACTORY TRAINING PROGRESS

6. Every effort must be made to assist a personnel having difficulty before it becomes necessary to convene a Training Review Board (TRB). Candidates who fail the Performance Check will be permitted one supplementary attempt.

COURSE REPORTS

7. In accordance with CFAO 26-12, a CF 377 will be completed for all candidates attending this course.

8. The “Pass: annotation may be expanded to reflect the ease with which a member achieved the course objectives if the qualification itself is inadequate for career

management or duty assignment purposes. In such instances, the following criteria must be applied:

a. met the standard with difficulty;
b. met the standard; or
c. exceeded the standard.

9. A narrative description of the member's course performance will be included in the CF 377 and will be completed as described in A-PM-245-001/FP-001.

TRAINING REVIEW BOARD

10. All Training Review Board requirements are contained in MARCORD 9-60 Volume 1.

CHAPTER 4: LESSON SPECIFICATIONS

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY THE COMMANDANT OF THE CANADIAN FORCES NAVAL ENGINEERING SCHOOL HALIFAX (CFNESH)

PO 401 MAINTAIN CPF DIESEL ENGINES

EO 401.1

1. Performance. Identify MWM Diesel Engine Components.

2. Conditions.

   a. Given:

      (1) operational or static MWM diesel engine;

      (2) MWM 226 static diesel mock-up;

(3) component display; and

(4) references.

b. Denied: N/A.

c. Environmental: N/A.

3. Standard.

a. Correctly identify MWM diesel engine components, including:

(1) equipment fitted on the engine;

(2) crankcase;

(3) cylinder liners;

(4) main bearing supports;

(5) crankshaft;

(6) connecting rods;

(7) main and connecting rod bearings;

(8) engine piston and piston ring arrangement;

(9) fly wheel;

(10) vibration damper;

(11) camshaft;

(12) gear train;

(13) cylinder head;

(14) injection pumps and injectors;

(15) speed governor; and

(16) turbocharger.

b. Explain the operation and operating parameters of the MWM 226 diesel engine; and

c. Explain the function of the MWM 226 diesel engine major components and sub-systems.

4. Time.

a. 8 x 50 minute periods for instruction.

5. Method/Media. Lecture.

6. References. A1, A5.

7. Training Aids.

a. LCD projector;

b. OHPs;

c. Handouts;

d. Movies/visual aids; and

e. Individual components.

8. Test Details.

a. The written EC will be administered in conjunction with EO 401.2.

b. The pass mark will be 70% and the EC will be worth 5% of the final mark for the course.

c. The material covering MWM 226 will not be tested.

9. Remarks. N/A.

10. Teaching Points,

a. Technical data:

(1) description of model number TBD 602 V16 K;

(2) type of design;

(3) firing technique;

(4) compression ratio;

(5) direction of rotation;

(6) cylinder numbering;

(7) firing sequence;

(8) valve timing procedure and clearances;

(9) injection, commencement of delivery;

(10) torque charts; and

(11) operating values.

b. Specialty tools:

(1) types; and

(2) function.

c. Fitted equipment:

(1) equipment fitted on the right hand side of the engine;

(2) equipment fitted on the left hand side of the engine;

(3) equipment fitted in the "V space of the engine; and

(4) equipment fitted on the front of the engine.

d. Crankcase: (1) layout;

(2) bearings;

(3) cooling water passages;

(4) removable wet cylinder liners;

(5) oil passages for piston cooling; and

(6) lubricating oil channels on the flywheel end.

e. Cylinder liners:

(1) measurement of wear;

(2) removal of liner;

(3) fitting of liner;

(4) measurement of head gasket; and

(5) replacing liner.

f. Mean bearing supports:

(1) numbering of bearings;

(2) thrust bearing; (3) torquing of bearing; and

(4) measuring of bearing.

g. Components attached to the crankshaft:

(1) counter weights;

(2) lubrication of connecting rod;

(3) fly wheel;

(4) vibration damper;

(5) drive gear; and

(6) removal of crankshaft.

h. Connecting rods:

(1) purpose;

(2) design;

(3) material;

(4) measurement of connecting rod big-end bearing; and

(5) purpose of the notching.

i. Main and connecting rod bearings:

(1) general;

(2) big end bearing; (3) crankshaft main bearing; and

(4) crankshaft thrust bearing.

j. Engine piston and piston ring arrangement:

(1) material and design;

(2) cooling;

(3) removal of piston and connecting rod;

(4) replacing piston and connecting rod; and

(5) replacing piston rings.

k. Flywheel:

(1) design; and

(2) purpose.

l. Vibration damper:

(1) design; and

(2) purpose.ts, including:

m. Camshaft:

(1) location;

(2) purpose;

(3) removal of camshaft; and

(4) removal of push rods.

n. Gear train:

(1) arrangement; and

(2) setting of gear markings.

o. Cylinder head:

(1) purpose;

(2) design;

(3) assembly of cylinder head;

(4) removal of cylinder head;

(5) replacing cylinder head;

(6) proper torque sequence;

(7) equalizing of valve bridges; and

(8) setting of rocker arms.

p. Injection valve:

(1) design; and

(2) operation.

q. Speed governor:

(1) design; and

(2) arrangement.

r. Turbocharger:

(1) purpose; and

(2) design.

s. MWM 226 diesel engine:

(1) operating principles;

(2) operating parameters;

(3) type of governor;

(4) operation of the governor;

(5) the fuel oil system;

(6) the lube oil system;

(7) the cooling system; (8) crankshaft and bearings; and

(9) liners.

PO 401 MAINTAIN CPF DIESEL ENGINES

EO 401.2

1. Performance. Identify MWM Diesel Engine Sub-Systems.

2. Conditions.

b. Given:

(1) operational or static MWM diesel engine; and

(2) references.

b. Denied: N/A.

c. Environmental: N/A.

3. Standard. Correctly identify MWM diesel engine sub-systems, including the:

a. lubricating oil system;

b. fuel system;

c. fresh water system;

d. salt water system;

e. starting air system; and

f. instrumentation and safety devices.

4. Time. 2 x 50 minute periods:

a. 1 x 50 minute period for instruction; and

b. 2 x 50 minute periods for written EC (covering EO 401.1 and EO 401.2).

5. Method/Media. Lecture.

6. References. A1.

7. Training Aids.

a. LCD projector;

b. OHPs;

c. Handouts; and

d. Visual aids..

8. Test Details.

a. There will be a written critical Enabling Check (EC) covering material in EO 401.1 and EO 401.2;

b. The pass mark is 70%;

c. The EC will constitute 5% of the final mark for the course; and

d. The material covering MWM 226 diesel engines will not be tested.

9. Remarks. N/A.

10. Teaching Points,

a. Lubricating oil system:

(1) description;

(2) components;

(3) oil pumps:

(a) main;

(b) pre-lube; and

(c) hand.

(4) piston cooling nozzles.

b. Fuel system:

(1) description;

(2) drive for injection pumps; (3) injection pump description; and

(4) injection pump timing.

c. Fresh water system:

(1) description;

(2) fresh water cooling pump;

(3) fresh water cooling heat exchanger; (4) fresh water cooling thermostatic valve; and

(5) fresh water cooling pre-heating.

d. Salt water system:

(1) description;

(2) salt water cooling pumps;

(3) inter-cooler; and

(4) salt water cooling thermostatic valve.

e. Starting air system:

(1) starter motor; and

(2) starter operation.

f. Instrumentation and safety devices:

(1) lube oil pressure switches, temperature and level transmitter;

(2) fuel leakage switch;

(3) filling limitation;

(4) fresh water sensors;

(5) RPM transmitter; and

(6) exhaust gas temperature transmitter.

4-12

CRITICAL ENABLING CHECK - EO 401.1 and EO 401.2

PART I – ADMINISTRATIVE INSTRUCTIONS

MATERIALS AND FACILITIES REQUIRED

1. Classroom facilities and writing materials.

ARRANGEMENTS AND PROCEDURES

2. Any trainee caught using any material that has not been supplied by the exam invigilator in order to assist in the completion of the exam will immediately forfeit the exam. These events shall be reported to your immediate supervisor.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. You will have 1 x 50 minute period to complete this EC, which will cover material from EO 401.1 and EO 401.2.

2. You must attain a minimum overall pass mark of 70% on this EC, which constitutes 5% of the final mark for the course.

3. References are not permitted.

4. You may ask for clarification of any question. Raise your hand and the instructor will come to you.

5. You are not to communicate with other students at any time during the examination period.

6. When you have completed your examination, raise your hand and the instructor will collect your paper. You will then leave the room quietly and make yourself available until the completion of the exam period.

7. The use of any material to assist in the completion of the exam that has not been supplied by the exam invigilator will constitute an immediate forfeiture of the exam.

PART III – ENABLING CHECK NOTES

1. This is a knowledge EC based on questions taken from the exam bank.

4-13

PO 401 MAINTAIN CPF DIESEL ENGINES.

EO 401.3

1. Performance. Perform MWM Diesel Engine Maintenance.

2. Conditions.

c. Given:

(1) operational or static MWM diesel engine;

(2) MWM 226 static diesel mock-up;

(3) iInjector tester;

(4) valve spring tester;

(5) oil test kit;

(6) NSN: 5180-12-189-6685 tool box;

(7) NSN: 5180-12-189-4431 tool box;

(8) specialized tools for MWM diesel engines;

(9) measuring devices;

(10) torque wrenches; (11) replacement parts; and

(12) references.

b. Denied: N/A.

c. Environmental: N/A.

3. Standard. Adhering to all safety precautions and procedures, and using the appropriate tools and test equipment, correctly perform MWM diesel engine maintenance, including:

a. preventive maintenance;

b. engine disassembly;

c. engine re-assembly; and

d. MWM 226 diesel engine maintenance and tune-up procedures.

4. Time. 39 x 50 minute periods:

a. 31 x 50 minute period for instruction and practice; and

b. 8 x 50 minute periods for practical Performance Check.

5. Method/Media. Lecture/demonstration/practice.

6. References. A1, A2, A5, B1.

7. Training Aids. N/A.

8. Test Details.

a. A Checklist will be used as the marking tool for the Performance Check, and the marking scheme will be as follows:

(1) Perform preventive maintenance 10%

(2) Disassemble the MWM diesel engine sub-system 15%

(3) Disassemble cylinders 15%

(4) Reassemble cylinders 15%

(5) Set gear train timing marks 15%

(6) Set injection pump timing 15%

(7) Assemble engine sub-system 15%

b. The PC will be worth 30% of the final mark for the course; and

c. The MWM 226 diesel engine maintenance and tune-up procedures will not be tested.

4-15

9. Remarks.

a. This EO is instructed as a team activity and the Performance Check will be administered as a team activity;

b. The instructor will maintain a checklist for each trainee and trainees will be assessed based on technical ability, class participation, enthusiasm and general knowledge in performing critical maintenance tasks; and

c. The engine must turn over freely after maintenance has been completed in order for the class to achieve a passing mark.

10. Teaching Points,

a. Perform preventive maintenance:

(1) perform scheduled PM tasks;

(2) maintain engine fluid systems; and

(3) conduct periodic engine performance test.

b. Disassemble the MWM diesel engine sub-system:

(1) remove air intercooler and associated piping;

(2) remove crankcase ventilation system;

(3) remove high pressure fuel lines;

(4) remove intercooler support stand;

(5) remove exhaust covers and manifolds;

(6) remove fresh water return manifold;

(7) remove fuel return and oil supply manifolds (together);

(8) remove air intake manifolds;

(9) remove oil return manifolds;

(10) remove fuel supply and return lines from the fuel injection pumps;

(11) remove starter;

(12) break coupling between both injection pumps;

(13) remove #2 injection pump;

(14) remove filling reduction and stop solenoid;

(15) break coupling between #1 injection pump and drive gear;

(16) remove #1 injection pump;

(17) remove injection pump coupling flange;

(18) remove oil supply and return lines from both turbochargers; and

(19) disassemble lubricating oil centrifuge.

c. Disassemble cylinders:

(1) remove all valve covers;

(2) losen all rocker gear bolts;

(3) remove rocker gear and push rods;

(4) remove injectors;

(5) loosen head holts;

(6) remove cylinder heads;

(7) remove crankcase doors and interference gear, if required;

(8) remove piston oil cooling nozzle;

(9) remove piston and connecting rod assemblies;

(10) remove piston rings and piston pins;

(11) remove cylinder liners; and (12) clean all dismantled parts.

4-17

d. Reassemble cylinders:

(1) lap in cylinder liners;

(2) install liners;

(3) check protrusion of liners and sealing rings;

(4) assemble connecting rod to piston;

(5) install piston and connecting rod assembly;

(6) install cooling water ferrules;

(7) install cylinder heads;

(8) install injector and injector clamps;

(9) loosely install rocker gear and push rods; and

(10) install piston cooling nozzle.

e. Set gear train timing marks:

(1) remove flywheel;

(2) remove gear train cover;

(3) remove the step idler;

(4) line up all timing marks and install idler gear;

(5) install gear train cover; and

(6) install flywheel.

f. Set injection pump timing:

(1) position flywheel at proper degree prior TDC #1;

(2) disconnect both couplings to injection pumps;

(3) set timing of #1 injection pump;

(4) reassemble coupling between drive gear and #1 injection pump;

(5) position flywheel at proper degree prior TDC #5;

(6) install and time #2 injection pump;

(7) assemble coupling between both injection pumps;

(8) set timing of both injection pumps together;

(9) install fuel rack indicator;

(10) install and adjust the filling reduction solenoid;

(11) adjust governor linkage; and

(12) install stop solenoid.

g. Assemble engine sub-system:

(1) install oil supply and return line from both turbochargers;

(2) install starter;

(3) install fuel supply and return lines to fuel both injection pumps;

(4) torque all rocker gear;

(5) equalize valve bridges;

(6) set valve characteristics;

(7) install valve covers;

(8) install oil return manifolds;

(9) install air intake manifolds;

(10) install fuel return and oil supply manifolds;

(11) install fresh water return manifolds;

(12) install exhaust manifolds and covers;

(13) install intercooler support stand;

(14) install high pressure fuel lines; (15) install crankcase ventilation system; and (16) install intercooler and associated piping.

h. MWM 226 diesel engine tune-up practices and procedures:

(1) remove, inspect and replace cylinder head;

(2) remove, inspect and replace piston and connecting rod;

(3) remove, inspect and replace cylinder liner;

(4) adjust valve clearances; and

(5) set commencement of fuel delivery.

PERFORMANCE CHECK EO 401.3

PART I – ADMINISTRATIVE INSTRUCTIONS

MATERIALS AND FACILITIES REQUIRED

1. One checklist per trainee.

2. One technician's toolbox per class.

3. One static MWM or Pielstick engine per class.

4. One set of special tools per class.

5. Space to conduct diesel maintenance.

ARRANGEMENTS AND PROCEDURES

6. This Performance Check will be completed as a team, although members will be assessed individually by the instructor by means of individual checklists.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. You will be assessed individually, although you will be working as part of a team.

2. You will have 8 x 50 minute periods to complete this Performance Check.

3. The PC cannot be passed unless successful maintenance is completed by the class as a team. if the engine cannot be barred over freely on completion, it will be considered as a failure.

PART III – PERFORMANCE CHECK NOTES

1. During this PC, two power packs will be removed from the diesel engine. The power packs will be replaced and the engine timing set by the class as a whole.

2. Individual members will be assessed via individual checklists held by the instructor.

3. The engine must be able to rotate on completion of maintenance in order for the class to attain a passing mark.

4. This PC is worth 30% of the final mark for PO 401.

4-21

PART IV - PERFORMANCE CHECKLIST

Name:

SN:

Critical Actions Scoring: 1 = Low 5 = High	Participation	Enthusiasm	Technical Ability	General Knowledge	Comments
Remove/replace cylinder head					
Remove/replace piston					
Remove/replace connecting rod					
Remove/replace liner					
Check component clearances against specifications					

Set commencement of fuel delivery					
Set valve adjustment					

PO 401 MAINTAIN CPF DIESEL ENGINES

EO 401.4

1. Performance. Identify Pielstick Diesel Engine Components.

2. Conditions.

   a. Given:

      (1) operational or static Pielstick diesel engine;

      (2) component display; and

      (3) references.

   b. Denied: N/A.

   c. Environmental: N/A.

3. Standard. Correctly identify Pielstick diesel engine components, including:

      (1) equipment fitted on the engine;

      (2) crankcase;

      (3) cylinder liners;

      (4) main bearing supports;

      (5) crankshaft;

      (6) connecting rods;

(7) main and connecting rod bearings;

(8) engine piston and piston ring arrangement;

(9) flywheel;

(10) vibration damper;

(11) camshaft;

(12) gear train;

(13) cylinder head;

(14) injection pumps and injectors;

(15) hydraulic and electronic speed governors; and

(16) turbochargers.

4. Time. 2 x 50 minute periods for instruction.

5. Method/Media. Lecture.

6. References. A3.

7. Training Aids.

a. LCD projector;

b. OHPs; and

c. Handouts.

8. Test Details. There will be a written critical Enabling Check (EC) administered in combination with EO 401.5.

9. Remarks. N/A.

10. Teaching Points,

a. Technical data:

(1) description of model number 20 PA6 V280 MPC;

(2) type of design;

(3) firing technique;

(4) compression ratio;

(5) direction of rotation;

(6) cylinder numbering;

(7) firing sequence;

(8) valve timing procedure and clearances;

(9) injection, commencement of delivery;

(10) torque charts; and

(11) operating values.

b. Specialty tools:

(1) types; and

(2) function.

c. Fitted equipment:

(1) equipment fitted on the right hand side of the engine;

(2) equipment fitted on the left hand side of the engine;

(3) equipment fitted on the 'TGOS' of the engine; and

(4) equipment fitted on the 'TGS' of the engine.

d. Crankcase:

(1) layout;

(2) cooling waster passages;

(3) oil passages;

(4) tie bolts and main bearing hydraulic bolting; and

(5) crankcase external equipment.

e. Cylinder liners:

(1) measurement of wear;

(2) removal of liner;

(3) fitting of liner;

(4) measurement of head gasket; and

(5) replacing liner.

f. Main bearings and thrust bearings:

(1) numbering of bearings;

(2) thrust bearing; (3) torquing of bearing; and

(4) measuring of bearing.

g. Crankshaft and attached components:

(1) counter weights;

(2) Geislinger coupling/flywheel;

(3) vibration damper;

(4) timing gear; and

(5) removal of crankshaft.

h. Connecting rods:

(1) purpose;

(2) design;

(3) material;

(4) measurement of connecting rod big end bearing; and (5) fasteners.

4-26

i. Engine piston and piston ring arrangement:

(1) material and design;

(2) cooling;

(3) removing piston and connecting rod; (4) replacing piston and connecting rod; and

(5) replaciong piston rings.

j. Cylinder head:

(1) purpose;

(2) design;

(3) assembly of cylinder head;

(4) removal of cylinder head;

(5) replacing cylinder head;

(6) proper torquing sequence;

(7) hydraulic tightener;

(8) setting valve bridges; and

(9) setting rocker arms.

k. Injection valve:

(1) design; and

(2) operation.

l. Speed governor:

(1) design; and

(2) arrangement.

4-27

m. Camshaft:

(1) location;

(2) purpose;

(3) removal of camshaft; and

(4) removal of push rods.

n. Gear train:

(1) location; and

(2) setting of gear markings.

o. Turbocharger:

(1) design; and

(2) purpose.

4-28

PO 401 MAINTAIN CPF DIESEL ENGINES

EO 401.5

1. Performance. Identify Pielstick Diesel Engine Sub-Systems.

2. Conditions.

   a. Given:

      (1) operational or static Pielstick diesel engine;

      (2) component display; and

      (3) references.

   b. Denied: N/A.

   c. Environmental: N/A.

3. Standard. Correctly identify Pielstick diesel engine sub-systems, including:

      (1) lubricating oil system;

      (2) fuel system;

      (3) intake air system;

      (4) fresh water system;

      (5) salt water system;

      (6) starting air system; and (7) instrumentation and safety devices.

4. Time. 2 x 50 minute periods for instruction:

   a. 1 x 50 minute period for instruction; and

   b. 1 x 50 minute period for written critical EC.

5. Method/Media. Lecture.

6. References. A3.

7. Training Aids.

   a. LCD projector;

   b. OHPs; and

   c. Handouts.

8. Test Details.

   a. A written critical Enabling Check (EC) will be administered, covering information from EO 401.4 and EO 401.5;

   b. The pass mark will be 70%; and

   c. The EC will constitute 5% of the final mark for the course.

9. Remarks. N/A.

10. Teaching Points,

    a. Lubricating oil system:

       (1) description;

       (2) components;

       (3) oil pumps: (a) main; and

           (b) pre-lube.

       (4) oil cooler.

    b. Fuel system:

       (1) description;

       (2) gear pump;

(3) injection pumps; and

(4) injectors.

c. Fresh water system:

(1) description;

(2) fresh water cooling pump;

(3) fresh water cooling heat exchanger; (4) fresh water cooling pre-heating; and

(5) inter-cooler.

d. Salt water system:

(1) description;

(2) salt water cooling pump; (3) salt water cooling thermostatic valve; and

(4) air conditioning unit.

e. Starting air system:

(1) description; (2) starting air distributor; and

(3) starter operation.

f. Instrumentation and safety devices:

(1) lube oil pressure monitoring unit;

(2) oil level transmitter;

(3) fuel leakage detection boxes;

(4) bypass valves (P1-P2, P2-P4, P3-P7);

(5) fresh water sensors;

(6) RPM magnetic pick-up;

(7) Scrutomat exhaust and bearing temperature monitor; and (8) engine safety monitor.

CRITICAL ENABLING CHECK - EO 401.4 AND EO 401.5

PART I – ADMINISTRATIVE INSTRUCTIONS

MATERIALS AND FACILITIES REQUIRED

1. Classroom facilities and writing materials.

ARRANGEMENTS AND PROCEDURES

2. Any trainee caught using any material that has not been supplied by the exam invigilator in order to assist in the completion of the exam will immediately forfeit the exam. These events shall be reported to your immediate supervisor.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. You will have 1 x 50 minute period to complete this EC, which will cover all material covered during EO 401.4 and EO 401.5..

2. You must attain a minimum overall pass mark of 70% on this EC, which counts towards 5% of the final mark for PO 401.

3. You may ask for clarification of any question. Raise your hand and the instructor will come to you.

4. You are not to communicate with other students at any time during the examination period.

5. When you have completed your examination, raise your hand and the instructor will collect your paper. You will then leave the room quietly and make yourself available until the completion of the exam period.

6. The use of any material to assist in the completion of the exam that has not been supplied by the exam invigilator will constitute an immediate forfeiture of the exam.

PART III – ENABLING CHECK NOTES

1. This is a knowledge EC based on questions taken from the exam bank and will cover EO 401.4 and EO 401.5.

PO 401 MAINTAIN CPF DIESEL ENGINES

EO 401.6

1. Performance. Perform Pielstick Diesel Engine Maintenance.

2. Conditions.

a. Given:

(1) static Pielstick diesel engine;

(2) component display;

(3) valve spring tester;

(4) specialized tools for Pielstick diesel engine;

(5) measuring devices;

(6) torque wrenches; (7) replacement parts; and

(8) references.

b. Denied: N/A.

c. Environmental: N/A.

3. Standard. Adhering to all safety regulations and procedures, and using the appropriate tools and test equipment, correctly perform maintenance on the Pielstick diesel engine, including:

a, preventive maintenance; b.

engine disassembly; and

c. engine reassembly.

4. Time. 35 x 50 minute periods for instruction:

   a. 27 x 50 minute periods for instruction/practice; and

   b. 8 x 50 minute periods for practical PC.

5. Method/Media. Demonstration/practice.

6. References. A3.

7. Training Aids.

   a. LCD projector;

   b. OHPs; and

   c. Handouts.

8. Test Details.

   a. A checklist will be used as the marking tool for the Performance Check, and the marking scheme will be as follows:

(1)	Perform preventive maintenance	10%
(2)	Disassemble the engine sub-system	15%
(3)	Disassemble cylinders	15%
(4)	Reassemble cylinders	15%
(5)	Set gear train timing marks	15%
(6)	Set injection pump timing	15%
(70	Assemble engine sub-system	15%

   b. The PC will be worth 30% of the final mark for the course.

   c. EO is instructed as a team activity and the Performance Check will be administered as a team activity;

   d. The instructor will maintain a checklist for each student and students will be assessed based on technical ability, class participation, enthusiasm and general knowledge in performing critical maintenance tasks; and

e. The engine must turn over freely after maintenance has been completed in order for the class to achieve a passing grade.

4-35

9. Remarks. N/A.

10. Teaching Points,

a. Perform preventive maintenance:

(1) perform scheduled PM tasks;

(2) maintain engine fluid systems; and

(3) conduct periodic engine performance test.

b. Disassemble the Pielstick diesel engine sub-system:

(1) prepare hydraulic ram head bolt removal kit;

(2) remove cylinder head cover;

(3) remove high pressure fuel lines;

(4) remove exhaust covers and manifolds;

(5) remove fresh water return manifold;

(6) remove fuel return and supply manifolds;

(7) remove air intake manifolds;

(8) remove oil return manifolds;

(9) remove high pressure fuel lines from the fuel injection pumps;

(10) disassemble lubricating oil centrifuge; and

(11) disassemble Moatti self-cleaning filter.

c. Disassemble cylinders:

(1) loosen all rocker gear bolts;

(2) remove rocker gear and push rods;

(3) remove injectors;

4-36

(4) fit hydraulic ram head bolt removal tool;

(5) loosen and remove head bolts;

(6) using special tool, remove cylinder head;

(7) disassemble cylinder head and examine components;

(8) remove crankcase doors and interference gear, if required;

(9) using special tool, remove big end bearing cap;

(10) using special tool, remove piston and connecting rod assembly;

(11) using special tool, remove piston rings and piston pins;

(12) disassemble piston;

(13) using special tool, remove cylinder liners; and

(14) clean and measure all dismantled parts.

d. Re-assemble cylinders, re-tighten to specific torque values:

(1) install liners;

(2) check protrusion of liners and sealing rings;

(3) assemble piston and connecting rod;

(4) install piston and connecting rod assembly;

(5) install big end bearing cap;

(6) re-assemble cylinder head;

(7) install cylinder head;

(8) using hydraulic tool, tighten cylinder head to specifications;

(9) install injector and injector clamps;

(10) install push rods and rocker gear; and

(11) adjust valve clearances.

e. Check gear train timing marks:

(1) remove access cover at TGS;

(2) install special camshaft timing tool;

(3) rotate engine, lining up all timing marks;

(4) remove timing tool; and

(5) install access cover.

f. Assemble engine sub-system:

(1) assemble and install Moatti filter;

(2) assemble lubricating oil centrifuge;

(3) install injection pump high pressure fuel lines;

(4) install lube oil jumper lines;

(5) install air intake;

(6) install fuel supply and return manifolds;

(7) install fresh water manifold;

(8) install exhaust manifolds and covers; and (9) install cylinder head cover.

PERFORMANCE CHECK EO 401.6

PART I – ADMINISTRATIVE INSTRUCTIONS

MATERIALS AND FACILITIES REQUIRED

1. One checklist per trainee.

2. One technician's toolbox per class.

3. One static MWM or Pielstick engine per class.

4. One set of special tools per class.

5. Space to conduct diesel maintenance.

ARRANGEMENTS AND PROCEDURES

6. This Performance Check will be completed as a team, although members will be assessed individually by the instructor by means of individual checklists.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. You will be assessed individually, although you will be working as part of a team.

2. You will have 8 x 50 minute periods to complete this Performance Check.

3. The PC cannot be passed unless successful maintenance is completed by the class as a team. if the engine cannot be barred over freely on completion, it will be considered as a failure.

PART III – PERFORMANCE CHECK NOTES

1. During this PC, two power packs will be removed from the diesel engine. The power packs will be replaced and the engine timing set by the class as a whole.

2. Individual members will be assessed via individual checklists held by the instructor.

3. The engine must be able to rotate on completion of maintenance in order for the class to attain a passing mark.

4. This PC is worth 30% of the final mark for PO 401.

4-39

PART IV - PERFORMANCE CHECKLIST

Name:

SN:

Critical Actions	Participation	Enthusiasm	Technical Ability	General Knowledge	Comments
Scoring: 1 = Low 5 = High					
Remove/replace cylinder head					
Remove/replace piston					
Remove/replace connecting rod					
Remove/replace liner					
Check component clearances against specifications					

Chapters 1,2:
Chapters 3,4:

Set commencement of fuel delivery					
Set valve adjustment					

PO 401 MAINTAIN CPF DIESEL ENGINES

EO 401.7

1. Performance. Troubleshoot Diesel Engines.

2. Conditions.

   a. Given:

      (1) operational MWM 602 diesel engine;

      (2) operational Volvo Penta diesel engine;

      (3) static Pielstick diesel engine;

      (4) troubleshooting charts;

      (5) induced discrepancies in engine parameters;

      (6) tools and test equipment; and

      (7) references.

   b. Denied: N/A.

   c. Environmental: N/A.

3. Standard. Adhering to all safety regulations and procedures, and using troubleshooting charts, troubleshoot the diesel engine by:

   a. solving problems; and

b. diagnosing discrepancies in engine parameters.

4. Time. 6 x 50 minute periods:

a. 2 x 50 minute periods for instruction/practice; and

b. 4 x 50 minute periods for practical PC.

5. Method/Media. Demonstration/practice.

4-41

6. References. A3, A5, C1.

7. Training Aids. N/A.

8. Test Details.

a. A checklist will be used as the marking tool for the Performance Check, and the marking scheme will be as follows:

(1)	Perform preventive maintenance	10%
(2)	Disassemble the engine sub-system	15%
(3)	Disassemble cylinders	15%
(4)	Reassemble cylinders	15%
(5)	Set gear train timing marks	15%
(6)	Set injection pump timing	15%
(70	Assemble engine sub-system	15%

b. The PC will be worth 30% of the final mark for the course.

c The PC will be performed individually. d The

instructor will maintain a checklist for each trainee.

e. Trainees will be provided with five troubleshooting scenarios and will be assessed on their ability to correctly identify the likely cause of each fault and the corrective action to be taken.

f. Three of the five scenarios must be correctly identified in order for trainees to pass the PC.

9. Remarks. A generic troubleshooting phrase shall be conducted, utilizing operational MWM and Volvo Penta training engines as suitable substitutes. Trainees will be expected to investigate and solve specific faults.

10. Teaching Points,

a. Insufficient lube oil pressure;

b. Excessive cooling water temperature;

c. Excessive exhaust temperature;

d. Loss of power;

e. Engine does not start; and

f. High crankcase pressure.

PERFORMANCE CHECK EO 401.7

PART I – ADMINISTRATIVE INSTRUCTIONS

MATERIALS AND FACILITIES REQUIRED

1. One checklist per trainee.

2. One technician's toolbox per class.

3. One static Volvo engine per class.

4. One set of special tools per class.

5. Space to conduct diesel maintenance.

ARRANGEMENTS AND PROCEDURES

6. This Performance Check will be completed as a team, although members will be assessed individually by the instructor by means of individual checklists.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. You will be assessed individually, although you will be working as part of a team.

2. You will have 4 x 50 minute periods to complete this Performance Check.

3. You will be provided with five (5) troubleshooting scenarios on an operational Volvo diesel engine in the small test cell.

4. You will be assessed on your ability to correctly identify the likely cause of each fault and the corrective action to be taken.

5. Three of the five scenarios must be correctly identified to achieve a passing grade.

6. The faults will be the same for all trainees.

PART III – PERFORMANCE CHECK NOTES

1. Trainees will be provided with five (5) troubleshooting scenarios to be assessed individually by means of a checklist.

2. The faults the same for all trainees.

3. Three of the five faults must be correctly diagnosed in order to achieve a pass.

4. The Performance Check is worth 30% of the final mark for PO 401.

PART IV - PERFORMANCE CHECKLIST

Name:

SN:

Critical Actions Scoring: 1 = Low 5 = High	Participation	Enthusiasm	Technical Ability	General Knowledge	Comments
Fail to Start Start Relay					
Fail to Start Emergency Stop Linkage					
Lube Oil Pressure High or No Indication					
High FW Temperature High or No Indication					
High Crankcase Pressure High Manometer Reading					

Loss of Power Fuel Restriction					
Loss of Power Air Restriction					

4-45

ANNEX A – MAIN REFERENCES

REFERENCE	CONTROL NUMBER	TITLE
A1	C-26-411-000/MS-001	MWM Technical Manual
A2	C-26-411-000/MY-001	MWM Parts Manual
A3	C-24-548-000/MS-001	Pielstick Operation and Maintenance Manual
A4	C-24-548-000/MS-002	Pielstick Parts and Tooling Manual
A5	C-27-499-000/MS-001	MWM 226 Technical Manual
B1	Publication 6.226.0.006.7802	MWM 226 Repair Manual
B2	Publication 0297 5201 TD 226	Deutz Service and Operation Manual
C1	ISBN 0-87006-424-X	Bill Toboldt. Diesel Fundamentals, Service and Repair, 1983.

4-A-1 ANNEX B– Suggested Sequencing and Timings

PO/EO	OBJECTIVE	EO (Theory/ Practical)	EC	PC	Total
PO 401	**Maintain CPF Diesel Engines**				**94**
EO 401.1	Identify MWM Diesel Engine Components	8			8
EO 401.2	Identify MWM Diesel Engine Sub-Systems	1	1 (C)		2
EO 401.3	Perform MWM Diesel Engine Maintenance and Main Turbines	31		8	39

EO 401.4	Identify Pielstick Diesel Engine Components	2			2
EO 401.5	Identify Pielstick Disel Engine Sub-Systems	1	1 (C)		2
EO 401.6	Perform Pielstick Diesel Engine Maintenance	27		8	35
EO 401.7	Troubleshoot Diesel Engines	2		4	6